

MATH20250 HW6 Observations

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Problem-Specific Observations

Be sure to consult the debrief file on Canvas and the compiled example solutions from your fellow students.

LADW 5.2.3. Let $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ be an orthonormal basis in V .

(a) Prove that for any $\mathbf{x} = \sum_{k=1}^n \alpha_k \mathbf{v}_k$, $\mathbf{y} = \sum_{k=1}^n \beta_k \mathbf{v}_k$,

$$(\mathbf{x}, \mathbf{y}) = \sum_{k=1}^n \alpha_k \bar{\beta}_k.$$

By substitution and linearity of the inner product, we have

$$(\mathbf{x}, \mathbf{y}) = \left(\sum_{k=1}^n \alpha_k \mathbf{v}_k, \sum_{k=1}^n \beta_k \mathbf{v}_k \right) = \sum_{k=1}^n \left(\alpha_k \mathbf{v}_k, \sum_{j=1}^n \beta_j \mathbf{v}_j \right) = \sum_{k=1}^n \alpha_k \left(\mathbf{v}_k, \sum_{j=1}^n \beta_j \mathbf{v}_j \right).^1$$

Conjugate symmetry, properties of the complex conjugate, and linearity of the inner product then give us

$$\begin{aligned} (\mathbf{x}, \mathbf{y}) &= \sum_{k=1}^n \alpha_k \left(\mathbf{v}_k, \sum_{j=1}^n \beta_j \mathbf{v}_j \right) = \sum_{k=1}^n \alpha_k \overline{\left(\sum_{j=1}^n \beta_j \mathbf{v}_j, \mathbf{v}_k \right)} = \sum_{k=1}^n \alpha_k \overline{\left(\sum_{j=1}^n (\beta_j \mathbf{v}_j, \mathbf{v}_k) \right)} \\ &= \sum_{k=1}^n \alpha_k \overline{\left(\sum_{j=1}^n \beta_j (\mathbf{v}_j, \mathbf{v}_k) \right)} = \sum_{k=1}^n \alpha_k \left(\sum_{j=1}^n \overline{\beta_j (\mathbf{v}_j, \mathbf{v}_k)} \right) = \sum_{k=1}^n \alpha_k \left(\sum_{j=1}^n \bar{\beta}_j \cdot \overline{(\mathbf{v}_j, \mathbf{v}_k)} \right). \end{aligned}$$

Note that α_k is a constant with respect to the sum over j . So, by the distributive property and conjugate symmetry, we now have

$$(\mathbf{x}, \mathbf{y}) = \sum_{k=1}^n \alpha_k \left(\sum_{j=1}^n \bar{\beta}_j \cdot \overline{(\mathbf{v}_j, \mathbf{v}_k)} \right) = \sum_{k=1}^n \sum_{j=1}^n \left(\alpha_k \bar{\beta}_j \cdot \overline{(\mathbf{v}_j, \mathbf{v}_k)} \right) = \sum_{k=1}^n \sum_{j=1}^n (\alpha_k \bar{\beta}_j \cdot (\mathbf{v}_k, \mathbf{v}_j)).$$

¹I switch the indexing variable in the second argument from k to j to note that the sums are independent from each other and head off any confusion that my result from them having the same indexing variable.

Recall that an orthonormal basis $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ has the following properties:

1. For any $k \in \{1, \dots, n\}$, $\|\mathbf{v}_k\| = 1 \Rightarrow \|\mathbf{v}_k\|^2 = (\mathbf{v}_k, \mathbf{v}_k) = 1$.
2. For all $j, k \in \{1, \dots, n\}$ such that $j \neq k$, $(\mathbf{v}_j, \mathbf{v}_k) = 0$.

The double sum we derived above has n^2 terms, n of which arise when $j = k$. As such, we can collapse the sum and use the properties of an orthonormal basis to conclude the following:

$$(\mathbf{x}, \mathbf{y}) = \sum_{k=1}^n \sum_{j=1}^n (\alpha_k \bar{\beta}_j \cdot (\mathbf{v}_k, \mathbf{v}_j)) = \sum_{k=1}^n (\alpha_k \bar{\beta}_k \cdot (\mathbf{v}_k, \mathbf{v}_k)) \Rightarrow (\mathbf{x}, \mathbf{y}) = \sum_{k=1}^n \alpha_k \bar{\beta}_k. \quad \square$$

(b) Deduce from this *Parseval's identity*

$$(\mathbf{x}, \mathbf{y}) = \sum_{k=1}^n (\mathbf{x}, \mathbf{v}_k) \overline{(\mathbf{y}, \mathbf{v}_k)}.$$

We begin by taking advantage of both the result from the previous part and the fact that, for all $k \in \{1, \dots, n\}$, $\|\mathbf{v}_k\| = 1 \Rightarrow \|\mathbf{v}_k\|^2 = 1$. So, we have

$$(\mathbf{x}, \mathbf{y}) = \sum_{k=1}^n \alpha_k \bar{\beta}_k = \sum_{k=1}^n \alpha_k \bar{\beta}_k \|\mathbf{v}_k\|^2 = \sum_{k=1}^n \alpha_k \bar{\beta}_k (\mathbf{v}_k, \mathbf{v}_k).$$

We now take advantage the fact that, for a fixed $k \in \{1, \dots, n\}$,

$$\sum_{j=1}^n (\alpha_j \mathbf{v}_j, \mathbf{v}_k) = (\alpha_k \mathbf{v}_k, \mathbf{v}_k).^2$$

Using that fact in conjunction with the commutativity of scalar multiplication and linearity of the inner product, we have

$$\begin{aligned} (\mathbf{x}, \mathbf{y}) &= \sum_{k=1}^n \alpha_k \bar{\beta}_k (\mathbf{v}_k, \mathbf{v}_k) = \sum_{k=1}^n \bar{\beta}_k \alpha_k (\mathbf{v}_k, \mathbf{v}_k) = \sum_{k=1}^n \bar{\beta}_k (\alpha_k \mathbf{v}_k, \mathbf{v}_k) = \sum_{k=1}^n \left(\bar{\beta}_k \sum_{j=1}^n (\alpha_j \mathbf{v}_j, \mathbf{v}_k) \right) \\ &= \sum_{k=1}^n \left(\bar{\beta}_k \left(\sum_{j=1}^n \alpha_j \mathbf{v}_j, \mathbf{v}_k \right) \right) = \sum_{k=1}^n \bar{\beta}_k (\mathbf{x}, \mathbf{v}_k). \end{aligned}$$

Employing a similar strategy, though now using conjugate symmetry where needed, we arrive at the following:

$$\begin{aligned} (\mathbf{x}, \mathbf{y}) &= \sum_{k=1}^n \bar{\beta}_k (\mathbf{x}, \mathbf{v}_k) = \sum_{k=1}^n \bar{\beta}_k (\mathbf{x}, \mathbf{v}_k) \|\mathbf{v}_k\|^2 = \sum_{k=1}^n \bar{\beta}_k (\mathbf{x}, \mathbf{v}_k) (\mathbf{v}_k, \mathbf{v}_k) = \sum_{k=1}^n (\mathbf{x}, \mathbf{v}_k) \bar{\beta}_k (\mathbf{v}_k, \mathbf{v}_k) \\ &= \sum_{k=1}^n (\mathbf{x}, \mathbf{v}_k) \bar{\beta}_k \cdot \overline{(\mathbf{v}_k, \mathbf{v}_k)} = \sum_{k=1}^n (\mathbf{x}, \mathbf{v}_k) \overline{\beta_k (\mathbf{v}_k, \mathbf{v}_k)} = \sum_{k=1}^n (\mathbf{x}, \mathbf{v}_k) \overline{(\beta_k \mathbf{v}_k, \mathbf{v}_k)} \\ &= \sum_{k=1}^n \left((\mathbf{x}, \mathbf{v}_k) \sum_{j=1}^n \overline{(\beta_j \mathbf{v}_j, \mathbf{v}_k)} \right) = \sum_{k=1}^n \left((\mathbf{x}, \mathbf{v}_k) \overline{\left(\sum_{j=1}^n \beta_j \mathbf{v}_j, \mathbf{v}_k \right)} \right) \Rightarrow (\mathbf{x}, \mathbf{y}) = \sum_{k=1}^n (\mathbf{x}, \mathbf{v}_k) \overline{(\mathbf{y}, \mathbf{v}_k)}. \quad \square \end{aligned}$$

²This follows from the pairwise orthogonality of orthonormal vectors.

(c) Assume now that $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ is only an orthogonal basis, not an orthonormal one. Can you write down Parseval's identity in this case?

If $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ is an orthogonal basis, we still have that the basis vectors are pairwise orthogonal to each other. However, we do not necessarily have that $\|\mathbf{v}_k\| = 1$ for all $k \in \{1, \dots, n\}$.

We can remedy this easily by normalizing each instance of $\mathbf{v}_k$, essentially recreating the conditions of the previous two parts. Therefore, for orthogonal $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$, we have

$$(\mathbf{x}, \mathbf{y}) = \sum_{k=1}^n \left(\mathbf{x}, \frac{\mathbf{v}_k}{\|\mathbf{v}_k\|} \right) \overline{\left(\mathbf{y}, \frac{\mathbf{v}_k}{\|\mathbf{v}_k\|} \right)} = \sum_{k=1}^n \frac{(\mathbf{x}, \mathbf{v}_k)}{\|\mathbf{v}_k\|} \overline{\frac{(\mathbf{y}, \mathbf{v}_k)}{\|\mathbf{v}_k\|}} \Rightarrow \boxed{(\mathbf{x}, \mathbf{y}) = \sum_{k=1}^n \frac{(\mathbf{x}, \mathbf{v}_k) \overline{(\mathbf{y}, \mathbf{v}_k)}}{\|\mathbf{v}_k\|^2}}.$$

LADW 5.3.2. Apply Gram-Schmidt orthogonalization to the system of vectors $(1, 2, 3)^T$, $(1, 3, 1)^T$. Write the matrix of the orthogonal projection onto the 2-dimensional subspace spanned by these vectors.

Remember that the goal with Gram-Schmidt orthogonalization is to find an orthogonal basis for the subspace spanned by non-orthogonal vectors, here $(1, 2, 3)^T$ and $(1, 3, 1)^T$. To begin the Gram-Schmidt algorithm, we start with

$$\mathbf{x} := \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}.$$

Let $E_{\mathbf{x}} = \text{span}(\{\mathbf{x}\})$. Given a vector $\mathbf{v}$, we want to find $\mathbf{y}$ such that $(\mathbf{y}, \mathbf{x}) = 0$. Recall that

$$\mathbf{x} \perp (\mathbf{v} - P_{E_{\mathbf{x}}} \mathbf{v}) \Rightarrow \mathbf{x} \perp \left(\mathbf{v} - \frac{(\mathbf{v}, \mathbf{x})}{\|\mathbf{x}\|^2} \mathbf{x} \right).$$

So, taking $\mathbf{v} = (1, 3, 1)^T$, we have

$$\mathbf{v} - \frac{(\mathbf{v}, \mathbf{x})}{\|\mathbf{x}\|^2} \mathbf{x} = \begin{pmatrix} 1 \\ 3 \\ 1 \end{pmatrix} - \frac{1(1) + 3(2) + 1(3)}{1^2 + 2^2 + 3^2} \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 1 \\ 3 \\ 1 \end{pmatrix} - \frac{5}{7} \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} = \frac{1}{7} \begin{pmatrix} 2 \\ 11 \\ -8 \end{pmatrix}.$$

Remember multiplying vectors by nonzero scalars does not affect their orthogonality to other vectors; as such, we will use

$$\mathbf{y} = \begin{pmatrix} 2 \\ 11 \\ -8 \end{pmatrix}.$$

So, after applying Gram-Schmidt, we have that

$$\text{span} \left(\left\{ \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \begin{pmatrix} 1 \\ 3 \\ 1 \end{pmatrix} \right\} \right) = \text{span} \left(\left\{ \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \begin{pmatrix} 2 \\ 11 \\ -8 \end{pmatrix} \right\} \right).$$

³Leaving fractions out of the orthogonalized basis is merely a matter of personal preference at this point. Later on, however, this choice will not be available.

Indeed, we may check and see that the standard inner product of our two vectors is, in fact, 0.

Let $E = \text{span}(\{\mathbf{x}, \mathbf{y}\})$. We now want to find the matrix A such that, for any $\mathbf{v} \in \mathbb{R}^3$,

$$A\mathbf{v} = P_E\mathbf{v}.$$

That is, if $E_{\mathbf{y}} = \text{span}(\{\mathbf{y}\})$, we need

$$A\mathbf{v} = (P_{E_{\mathbf{x}}} \mathbf{v} + P_{E_{\mathbf{y}}} \mathbf{v}),$$

using the fact that

$$P_E\mathbf{v} = \frac{(\mathbf{v}, \mathbf{x})}{\|\mathbf{x}\|^2} \mathbf{x} + \frac{(\mathbf{v}, \mathbf{y})}{\|\mathbf{y}\|^2} \mathbf{y}.$$

Additionally, for any $\mathbf{a} = (a_1, a_2, a_3)^T$, $\mathbf{b} = (b_1, b_2, b_3)^T$, recall that

$$(\mathbf{a}, \mathbf{b}) = \mathbf{b}^T \mathbf{a} = \mathbf{a}^T \mathbf{b} = b_1 a_1 + b_2 a_2 + b_3 a_3 = a_1 b_1 + a_2 b_2 + a_3 b_3.^4$$

NOTE: The following method of solving this problem is meant to give one “under the hood” look at what is happening, and it is **very inefficient**.

Please consult the exemplary solutions for more efficient methods.

For generic $\mathbf{x} = (x_1, x_2, x_3)^T$, $\mathbf{y} = (y_1, y_2, y_3)^T$, $\mathbf{v} = (v_1, v_2, v_3)^T \in \mathbb{R}^3$, we have that

$$A\mathbf{v} = P_{E_{\mathbf{x}}} \mathbf{v} + P_{E_{\mathbf{y}}} \mathbf{v} = \begin{pmatrix} \frac{(\mathbf{v}, \mathbf{x})}{\|\mathbf{x}\|^2} x_1 + \frac{(\mathbf{v}, \mathbf{y})}{\|\mathbf{y}\|^2} y_1 \\ \frac{(\mathbf{v}, \mathbf{x})}{\|\mathbf{x}\|^2} x_2 + \frac{(\mathbf{v}, \mathbf{y})}{\|\mathbf{y}\|^2} y_2 \\ \frac{(\mathbf{v}, \mathbf{x})}{\|\mathbf{x}\|^2} x_3 + \frac{(\mathbf{v}, \mathbf{y})}{\|\mathbf{y}\|^2} y_3 \end{pmatrix} = \begin{pmatrix} \frac{v_1 x_1 + v_2 x_2 + v_3 x_3}{\|\mathbf{x}\|^2} x_1 + \frac{v_1 y_1 + v_2 y_2 + v_3 y_3}{\|\mathbf{y}\|^2} y_1 \\ \frac{v_1 x_1 + v_2 x_2 + v_3 x_3}{\|\mathbf{x}\|^2} x_2 + \frac{v_1 y_1 + v_2 y_2 + v_3 y_3}{\|\mathbf{y}\|^2} y_2 \\ \frac{v_1 x_1 + v_2 x_2 + v_3 x_3}{\|\mathbf{x}\|^2} x_3 + \frac{v_1 y_1 + v_2 y_2 + v_3 y_3}{\|\mathbf{y}\|^2} y_3 \end{pmatrix}.$$

Redistributing, we observe that

$$\begin{aligned} A\mathbf{v} &= \begin{pmatrix} \frac{v_1 x_1 + v_2 x_2 + v_3 x_3}{\|\mathbf{x}\|^2} x_1 + \frac{v_1 y_1 + v_2 y_2 + v_3 y_3}{\|\mathbf{y}\|^2} y_1 \\ \frac{v_1 x_1 + v_2 x_2 + v_3 x_3}{\|\mathbf{x}\|^2} x_2 + \frac{v_1 y_1 + v_2 y_2 + v_3 y_3}{\|\mathbf{y}\|^2} y_2 \\ \frac{v_1 x_1 + v_2 x_2 + v_3 x_3}{\|\mathbf{x}\|^2} x_3 + \frac{v_1 y_1 + v_2 y_2 + v_3 y_3}{\|\mathbf{y}\|^2} y_3 \end{pmatrix} \\ &= \begin{pmatrix} \left(\frac{(x_1)^2}{\|\mathbf{x}\|^2} + \frac{(y_1)^2}{\|\mathbf{y}\|^2} \right) v_1 + \left(\frac{x_1 x_2}{\|\mathbf{x}\|^2} + \frac{y_1 y_2}{\|\mathbf{y}\|^2} \right) v_2 + \left(\frac{x_1 x_3}{\|\mathbf{x}\|^2} + \frac{y_1 y_3}{\|\mathbf{y}\|^2} \right) v_3 \\ \left(\frac{x_1 x_2}{\|\mathbf{x}\|^2} + \frac{y_1 y_2}{\|\mathbf{y}\|^2} \right) v_1 + \left(\frac{(x_2)^2}{\|\mathbf{x}\|^2} + \frac{(y_2)^2}{\|\mathbf{y}\|^2} \right) v_2 + \left(\frac{x_2 x_3}{\|\mathbf{x}\|^2} + \frac{y_2 y_3}{\|\mathbf{y}\|^2} \right) v_3 \\ \left(\frac{x_1 x_3}{\|\mathbf{x}\|^2} + \frac{y_1 y_3}{\|\mathbf{y}\|^2} \right) v_1 + \left(\frac{x_2 x_3}{\|\mathbf{x}\|^2} + \frac{y_2 y_3}{\|\mathbf{y}\|^2} \right) v_2 + \left(\frac{(x_3)^2}{\|\mathbf{x}\|^2} + \frac{(y_3)^2}{\|\mathbf{y}\|^2} \right) v_3 \end{pmatrix} \\ &= \begin{pmatrix} \left(\frac{(x_1)^2}{\|\mathbf{x}\|^2} + \frac{(y_1)^2}{\|\mathbf{y}\|^2} \right) + \left(\frac{x_1 x_2}{\|\mathbf{x}\|^2} + \frac{y_1 y_2}{\|\mathbf{y}\|^2} \right) + \left(\frac{x_1 x_3}{\|\mathbf{x}\|^2} + \frac{y_1 y_3}{\|\mathbf{y}\|^2} \right) \\ \left(\frac{x_1 x_2}{\|\mathbf{x}\|^2} + \frac{y_1 y_2}{\|\mathbf{y}\|^2} \right) + \left(\frac{(x_2)^2}{\|\mathbf{x}\|^2} + \frac{(y_2)^2}{\|\mathbf{y}\|^2} \right) + \left(\frac{x_2 x_3}{\|\mathbf{x}\|^2} + \frac{y_2 y_3}{\|\mathbf{y}\|^2} \right) \\ \left(\frac{x_1 x_3}{\|\mathbf{x}\|^2} + \frac{y_1 y_3}{\|\mathbf{y}\|^2} \right) + \left(\frac{x_2 x_3}{\|\mathbf{x}\|^2} + \frac{y_2 y_3}{\|\mathbf{y}\|^2} \right) + \left(\frac{(x_3)^2}{\|\mathbf{x}\|^2} + \frac{(y_3)^2}{\|\mathbf{y}\|^2} \right) \end{pmatrix} \mathbf{v} \\ &\Rightarrow A = \begin{pmatrix} \left(\frac{(x_1)^2}{\|\mathbf{x}\|^2} + \frac{(y_1)^2}{\|\mathbf{y}\|^2} \right) + \left(\frac{x_1 x_2}{\|\mathbf{x}\|^2} + \frac{y_1 y_2}{\|\mathbf{y}\|^2} \right) + \left(\frac{x_1 x_3}{\|\mathbf{x}\|^2} + \frac{y_1 y_3}{\|\mathbf{y}\|^2} \right) \\ \left(\frac{x_1 x_2}{\|\mathbf{x}\|^2} + \frac{y_1 y_2}{\|\mathbf{y}\|^2} \right) + \left(\frac{(x_2)^2}{\|\mathbf{x}\|^2} + \frac{(y_2)^2}{\|\mathbf{y}\|^2} \right) + \left(\frac{x_2 x_3}{\|\mathbf{x}\|^2} + \frac{y_2 y_3}{\|\mathbf{y}\|^2} \right) \\ \left(\frac{x_1 x_3}{\|\mathbf{x}\|^2} + \frac{y_1 y_3}{\|\mathbf{y}\|^2} \right) + \left(\frac{x_2 x_3}{\|\mathbf{x}\|^2} + \frac{y_2 y_3}{\|\mathbf{y}\|^2} \right) + \left(\frac{(x_3)^2}{\|\mathbf{x}\|^2} + \frac{(y_3)^2}{\|\mathbf{y}\|^2} \right) \end{pmatrix}. \end{aligned}$$

⁴N.B. The equation $(\mathbf{a}, \mathbf{b}) = \mathbf{b}^T \mathbf{a} = \mathbf{a}^T \mathbf{b}$ holds because $\mathbf{a}$ and $\mathbf{b}$ are elements of *real* vector spaces.

So, given $\|\mathbf{x}\|^2 = 14$ and $\|\mathbf{y}\|^2 = 189$, we have the following projection matrix A :

$$A = \begin{pmatrix} \left(\frac{1}{14} + \frac{4}{189}\right) & \left(\frac{2}{14} + \frac{22}{189}\right) & \left(\frac{3}{14} - \frac{16}{189}\right) \\ \left(\frac{2}{14} + \frac{22}{189}\right) & \left(\frac{4}{14} + \frac{121}{189}\right) & \left(\frac{6}{14} - \frac{88}{189}\right) \\ \left(\frac{3}{14} - \frac{16}{189}\right) & \left(\frac{6}{14} - \frac{88}{189}\right) & \left(\frac{9}{14} + \frac{64}{189}\right) \end{pmatrix} = \begin{pmatrix} \frac{27+8}{378} & \frac{54+44}{378} & \frac{81-32}{378} \\ \frac{54+44}{378} & \frac{108+242}{378} & \frac{162-176}{378} \\ \frac{81-32}{378} & \frac{162-176}{378} & \frac{243+128}{378} \end{pmatrix} = \begin{pmatrix} \frac{35}{378} & \frac{98}{378} & \frac{49}{378} \\ \frac{378}{378} & \frac{378}{350} & -\frac{14}{378} \\ \frac{378}{378} & -\frac{14}{378} & \frac{371}{378} \end{pmatrix}$$

$$\Rightarrow A = \begin{pmatrix} \frac{5}{54} & \frac{7}{27} & \frac{7}{54} \\ \frac{7}{27} & \frac{25}{27} & -\frac{1}{27} \\ \frac{7}{54} & -\frac{1}{27} & \frac{53}{54} \end{pmatrix} = \frac{1}{54} \begin{pmatrix} 5 & 14 & 7 \\ 14 & 50 & -2 \\ 7 & -2 & 53 \end{pmatrix}.$$

LADW 5.3.4. Find the distance from a vector $(2, 3, 1)^T$ to the subspace spanned by the vectors $(1, 2, 3)^T, (1, 3, 1)^T$. Note, that I am only asking to find the distance to the subspace, not the orthogonal projection.

The distance from a vector $\mathbf{v}$ to the subspace E is given by

$$\|\mathbf{v} - P_E \mathbf{v}\| = \|I\mathbf{v} - A\mathbf{v}\| = \|(I - A)\mathbf{v}\|,$$

where A is the matrix for the orthogonal projection of a vector onto E . So, using the matrix from the previous part and $\mathbf{v} = (2, 3, 1)^T$, we have the following:

$$\begin{aligned} \|\mathbf{v} - P_E \mathbf{v}\| &= \left\| \left(\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} - \frac{1}{54} \begin{pmatrix} 5 & 14 & 7 \\ 14 & 50 & -2 \\ 7 & -2 & 53 \end{pmatrix} \right) \begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix} \right\| \\ &= \left\| \frac{1}{54} \left(\begin{pmatrix} 54 & 0 & 0 \\ 0 & 54 & 0 \\ 0 & 0 & 54 \end{pmatrix} - \begin{pmatrix} 5 & 14 & 7 \\ 14 & 50 & -2 \\ 7 & -2 & 53 \end{pmatrix} \right) \begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix} \right\| = \frac{1}{54} \left\| \begin{pmatrix} 49 & -14 & -7 \\ -14 & 4 & 2 \\ -7 & 2 & 1 \end{pmatrix} \begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix} \right\| \\ &= \frac{1}{54} \left\| \begin{pmatrix} 98 - 42 - 7 \\ -28 + 12 + 2 \\ -14 + 6 + 1 \end{pmatrix} \right\| = \frac{1}{54} \left\| \begin{pmatrix} 49 \\ -14 \\ -7 \end{pmatrix} \right\| = \frac{7}{54} \left\| \begin{pmatrix} 7 \\ -2 \\ -1 \end{pmatrix} \right\| = \frac{7\sqrt{49 + 4 + 1}}{54} = \frac{7}{\sqrt{54}} \\ &\Rightarrow \|\mathbf{v} - P_E \mathbf{v}\| = \frac{7}{3\sqrt{6}} = \frac{7\sqrt{6}}{18}. \end{aligned}$$

LADW 6.2.7. True or false:

(a) A product of two self-adjoint matrices is self-adjoint.

Let A and B be self-adjoint $n \times n$ matrices; that is, $A = A^*$ and $B = B^*$. We then observe that

$$AB = A^* B^* = (BA)^*.$$

If we had the additional condition that $AB = BA$, we could conclude that the product of self-adjoint A and B is itself self-adjoint. However, since this is not the case in general, we have that this statement is **false**.

(b) If A is self-adjoint, then A^k is self-adjoint.

Let A be a self-adjoint $n \times n$ matrix; that is, $A = A^*$. Recall from the previous part that, for any two self-adjoint matrices A and B ,

$$AB = (BA)^* \Rightarrow AB = (AB)^* \text{ if } AB = BA.$$

Note that powers of the same matrix commute with each other, giving us

$$A = A^* \Rightarrow A^k = (A^k)^* \text{ for } k \in \mathbb{N} \cup \{0\}.$$

Therefore, this statement is **true**.

Supp. I. The standard dot product still makes sense in $(\mathbb{Z}/5\mathbb{Z})^2$. (Despite the fact that it's not clear what the “non-negativity” condition of an inner product would mean in this setting.) Call two vectors in $(\mathbb{Z}/5\mathbb{Z})^2$ orthogonal if their dot product is 0. How many vectors in $(\mathbb{Z}/5\mathbb{Z})^2$ are orthogonal to the vector $\begin{pmatrix} 2 \\ 1 \end{pmatrix}$? Does each non-zero vector in $(\mathbb{Z}/5\mathbb{Z})^2$ have the same number of vectors orthogonal to it, or not?

As usual, the set of vectors orthogonal to a collection of vectors in $(\mathbb{Z}/5\mathbb{Z})^2$ is the solution set of the homogeneous equation $A\mathbf{x} = \mathbf{0}$, where A is the matrix whose rows are the transposes of the vectors in the collection.

So, we have that the set of vectors orthogonal to $\begin{pmatrix} 2 \\ 1 \end{pmatrix}$ is every vector $\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$ where $x_1, x_2 \in \mathbb{Z}/5\mathbb{Z}$ satisfy

$$2x_1 = -x_2 \Leftrightarrow -2x_1 = x_2 \Leftrightarrow 3x_1 = x_2.$$

This gives us

$$\ker \left(\begin{pmatrix} 2 & 1 \end{pmatrix} \right) = \text{span} \left(\begin{pmatrix} 1 \\ 3 \end{pmatrix} \right) = \left\{ \begin{pmatrix} 1 \\ 3 \end{pmatrix}, \begin{pmatrix} 2 \\ 1 \end{pmatrix}, \begin{pmatrix} 3 \\ 4 \end{pmatrix}, \begin{pmatrix} 4 \\ 2 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \end{pmatrix} \right\}.$$

So, there are **5** vectors in $(\mathbb{Z}/5\mathbb{Z})^2$ that are orthogonal to $(2, 1)^T$.

Consider now all vectors in $(\mathbb{Z}/5\mathbb{Z})^2$ of the form $\begin{pmatrix} a \\ b \end{pmatrix}$, where at most one of a and b equals 0. We then observe that the solution set is $\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$, where $x_1, x_2 \in \mathbb{Z}/5\mathbb{Z}$ satisfy

$$ax_1 = -bx_2 \Leftrightarrow ax_1 = (5 - b)x_2.$$

If $a = 0$, then x_1 may be any of the 5 elements of $\mathbb{Z}/5\mathbb{Z}$, while x_2 must be 0; the opposite is true if $b = 0$. Otherwise, since $\mathbb{Z}/5\mathbb{Z}$ is a finite field with a prime order, every element has a unique additive inverse in $\mathbb{Z}/5\mathbb{Z}$ and every nonzero element has a unique multiplicative inverse in $\mathbb{Z}/5\mathbb{Z}$.

Consequently, if $x_1 \in \mathbb{Z}/5\mathbb{Z}$ is arbitrary, you will find $x_2 = ax_1(5 - b)^{-1}$; similarly, if $x_2 \in \mathbb{Z}/5\mathbb{Z}$ is arbitrary, $x_1 = a^{-1}(5 - b)x_2$. So, **all nonzero vectors in $(\mathbb{Z}/5\mathbb{Z})^2$ have 5 orthogonal vectors** in the space.

General Observations

IMPORTANT: Any problems with this boxed “x” next to them – **X** – should NOT be chosen for your HW submission. If you do this on one HW assignment, you will be warned but given credit. On subsequent assignments, you will not receive credit for problems that have been so marked.

- **Observations on Proofs**

- (a) **Directionality:** Suppose you have to prove “ A if and only if B .” ($A \Leftrightarrow B$) This is a **twofold** process:
 - i. **Prove “if A , then B .”** This may be done by supposing A is true and proving B directly, supposing A is true but B is false and proving by contradiction, or by contrapositive – “if A , then B ” is logically equivalent to “if not B , then not A .”
 - ii. **Prove “if B , then A .”** This may be done by supposing B is true and proving A directly, supposing B is true but A is false and proving by contradiction, or by contrapositive – “if B , then A ” is logically equivalent to “if not A , then not B .”
- (b) **Defining variables and other quantities:** Whenever you introduce a new quantity, be it a scalar, vector, etc., **state the space in which it lives.**

Ex. $\mathbf{v} \in V$ (where V is a vector space), $\alpha \in \mathbb{C}$.

The $\in$ symbol, meaning “is an element of,” is generated by `\in`.

- (c) **Existence:** If you are tasked with proving the existence of a quantity, **do not presuppose its existence in your notation.**
 - i. Example: LADW 1.6.7 – The goal is to prove that the matrix B is invertible given that the matrices A and AB are invertible. It may be tempting at some point in the proof to use the following fact from Theorem 6.3 (pg. 26):

$$(AB)^{-1} = B^{-1}A^{-1}.$$

However, there is a key condition attached to that fact:

If linear transformations A and B are **invertible**...

As such, this fact should not be employed unless you have already shown that A and B are invertible.

- (d) **Justifications:** There are two main things to remember when justifying the steps in a proof:
 - i. **Prove all facts on which you rely.** If you need to use a result that you did not already prove in class, **prove it explicitly.** If you need to use a result that comes from another homework problem, **prove it explicitly.** Do not take other homework problems for granted – see LADW 4.1.9.
 - ii. If you are unsure whether to justify something explicitly, **do it.** Better safe than sorry.

- Miscellaneous

- (a) **Know properties in full.** Unless you know that an inner product space is real, use *conjugate* symmetry of the inner product, not straight symmetry.
- (b) **Check your calculations.** Especially beware of dropping negative signs and accidentally performing the wrong arithmetic operation on two numbers (e.g., adding instead of multiplying).
 - i. When diagonalizing a matrix A , **supply the full $A = PDP^{-1}$ equation.** Additionally, make sure you order the eigenvalues in D to correspond with the proper eigenvectors in P .
 - ii. If you are finding a solution to a linear system of equations, **substitute your answer to check.** Sometimes, after converting to matrix form, it can be difficult to remember that there was an underlying group of equations. Use them to your advantage.
 - iii. If you are decomposing a matrix (e.g., writing it as a product of elementary matrices), **multiply the resulting matrices** to make sure you get the result you expect.
- (c) **Read the problems carefully.** Make sure you answer **every part of the problem.**
 - i. This also means you should not make any additional assumptions than what the problem allows.
 - ii. Additionally, for true/false questions, if a statement is true under a certain set of circumstances but *not under the set of circumstances outlined* in the statement, **the statement is false.** If no more specifics are given, err on the side of generality.